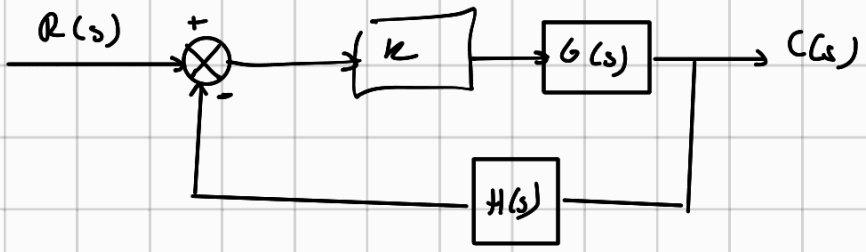
Root locus



$$\frac{C(s)}{R(s)} = \frac{k\,G(s)}{1 + k\,G(s)\,H(s)}$$

$$1 + k\,G(s)\,H(s) = 0 \Rightarrow G(s)\,H(s) = -\frac{1}{k}$$

- mag: $|G(s)H(s)| = \frac{1}{k}$
- phase: $\angle G(s)H(s) = \pm 180(2i+1)$, $i = 0, 1, \dots$

- $k = 0 \Rightarrow$ poles of the OL TF
- $k \uparrow \infty \rightarrow$ zeros " " "

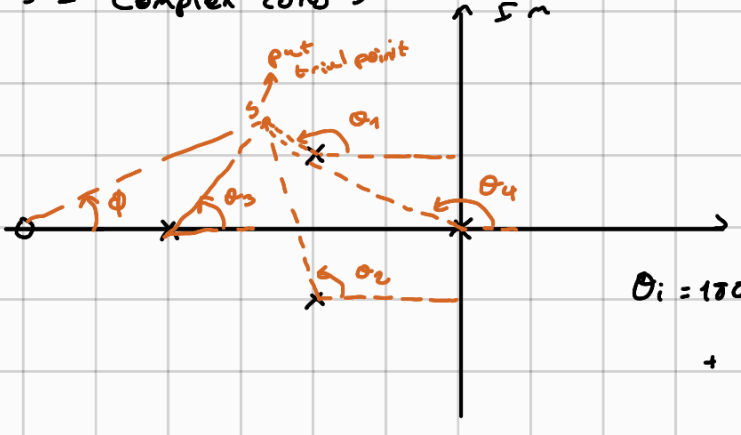
n: number of poles
m: " of zeros

- n: branches
- n − m: asymptotes

$$\phi = \pm \frac{180\,(2i+1)}{n-m}$$

Centroid: $\frac{\Sigma \text{Real poles} - \Sigma \text{Real zeros}}{n-m}$

s - Complex conjugate poles



$\angle$ trial point $= \phi_1 - \theta_1 - \theta_2 - \theta_3 - \theta_4 = 180$

↳ $\theta_1 = 180 - \phi_1 + \theta_2 + \theta_3 + \theta_4$

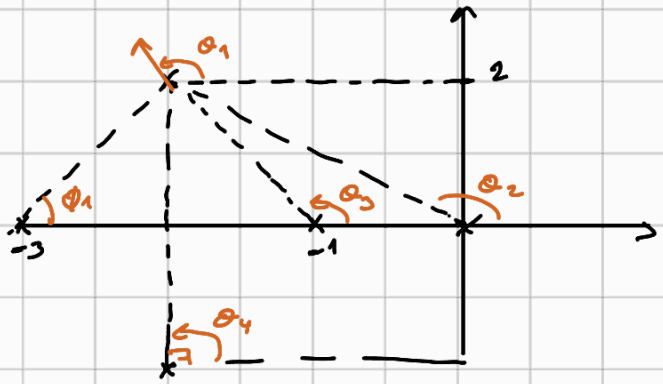
↓ departure angle from C.C pole

$\theta_i = 180 - \Sigma$ angles from the complex conjugate pole and other zero

$+ \Sigma$ angles from " " " poles

Departure angle from poles?

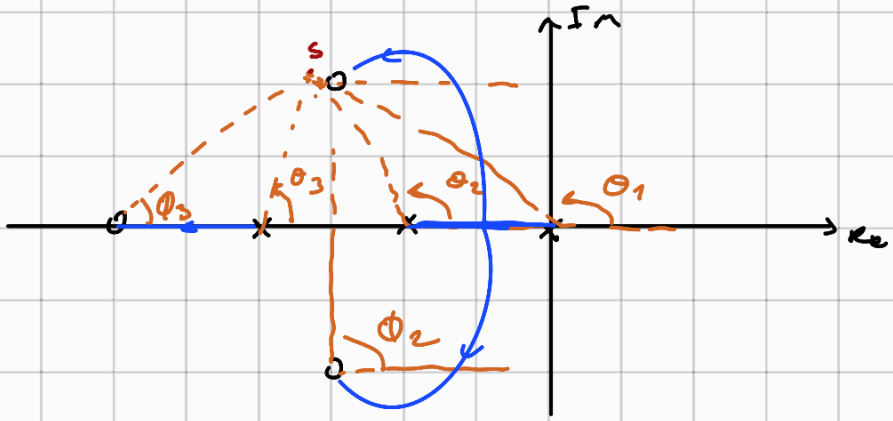
arrival angle " zeros



$$\theta_1 = 180 - \phi_1 + \theta_2 + \theta_3 + \theta_4$$

$$= 180 - \tan^{-1}\left(\frac{2}{2}\right) + \tan^{-1}\left(\frac{2}{-2}\right) + \tan^{-1}\left(\frac{2}{-1}\right) + 90^\circ$$

$\pi - \tan^{-1}\left(\frac{2}{2}\right)$



$$\phi_1 + \phi_2 + \phi_3 - \theta_1 - \theta_2 - \theta_3 = 180$$

$$\phi_1 = 180 - \phi_2 - \phi_3 + \theta_1 + \theta_2 + \theta_3$$

$$= 180 - \Sigma \text{angles from the trial point (cc. zero) to other zeros}$$

$$+ \Sigma \text{ " " " " " " poles}$$

Steps Intersection with the jw axis

I) $s = \sigma + jw$ subsitute $s = jw$ in the characteristic eq

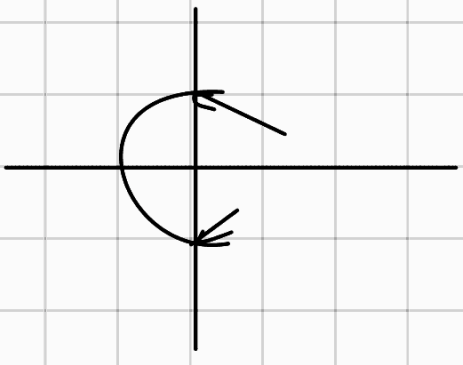
$$s^2 + 2s + 6 = 0$$

$$s^2 + 2s + k = 0$$

$$(jw)^2 + 2jw + k = 0$$

$$-w^2 + 2jw + k = 0 \Rightarrow \begin{cases} \text{Im}: & 2w = 0 \\ \text{Re}: & k - w^2 = 0 \end{cases} \Rightarrow \begin{cases} w = 0 \\ k = w^2 = 0 \end{cases}$$

$$s^3 + 2s^2 + 3s + 6 + k = 0$$

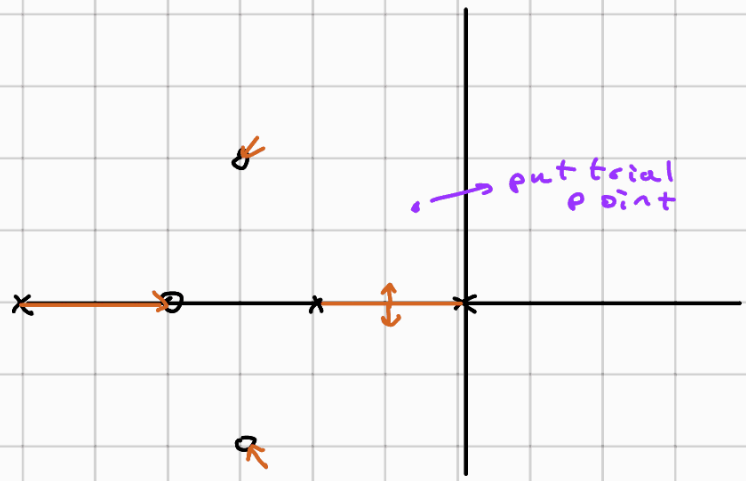


II) chacteric eq

routh table

if one of the components of the first column is zero

& make one full row zero!



or $s = \sigma + j\omega$

So Step 7 - Find the location or root locus of the complex plane!

Methde I) do numerial trial s error to discover the locations of 's' which statisfy the phase condition

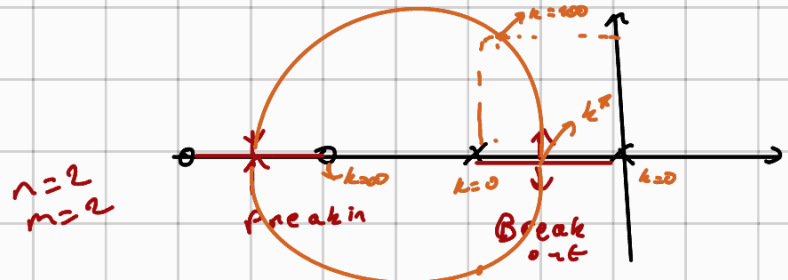
Methode II) Substitute s as $s = \sigma + j\omega$ in the characteristic equation and matematically find the phase condition.

$1 + k\,G(s)H(s) = 0$

$G(s)H(s) = -\frac{1}{k}$

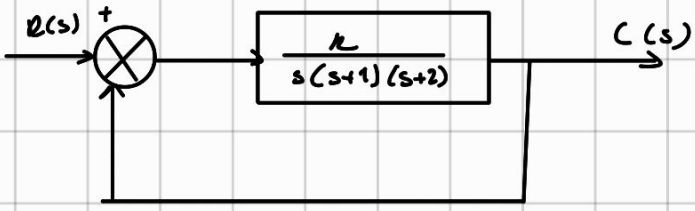
↳ mag = $|G(s)H(s)| = \frac{1}{k}$

↳ phase : $\angle G(s)H(s) = 180(2i+1)$



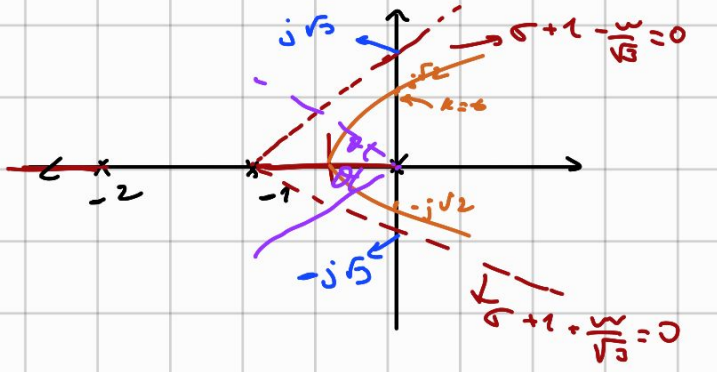
$$|G(s)H(s)| = \frac{\prod_{i=1}^{m} l_i}{\prod_{j=1}^{n} l_j}$$

l_i, l_j: magnitudes of individual complex function in numerator and denominator



$\angle G(s)H(s) = 180(2i+1)$

$-\angle s - \angle s+1 - \angle s+2 =$ total phase



- plot the root locus
- tune k so that $\zeta = 0.5$ for the dominant poles

$\begin{cases} n=3 \\ m=0 \end{cases}$ $\begin{cases} 3 \text{ branches} \\ 3 \text{ asymptotes} \end{cases}$

angles of asymptotes $= \frac{\mp 180(2i+1)}{n-m} = \mp 60(2i+1) \rightarrow 60, -60, 180$

Centroid: $\frac{\sum \text{real part of poles} - \sum \text{real parts of zeroes}}{n-m} = \frac{0-1-2-0}{3} = -1$

$s \to \infty \quad s \approx s+1 \approx s+2$

$$G(s) = \frac{k}{s(s+1)(s+2)} = \frac{k}{s^3+3s^2+2s} \simeq \frac{k}{s^3+3s^2+3s+1}$$

for large s values

Approximate $\bar{G}(s) \simeq \frac{k}{(s+1)^3}$

$\angle \bar{G}(s) = -3\angle s+1 = \mp 180(2i+1)$

$\Rightarrow \angle (s+1) = -60(2i+1)$

$s = \sigma + j\omega \Rightarrow \angle \sigma + j\omega + 1 = \tan^{-1}\left(\frac{\omega}{\sigma+1}\right) = \pm 60(2i+1) \rightarrow -60, 60, 180$

$\tan^{-1}\left(\frac{\omega}{\sigma+1}\right) = \pm 60, 180$

tan

$$\frac{w}{\sigma+1} = \begin{cases} \to \tan(60) = \sqrt{3} \\ \to \tan(-60) = -\sqrt{3} \\ \to \tan(180) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} w = \sqrt{3}(\sigma+1) \\ w = -\sqrt{3}(\sigma+1) \\ w = 0 \end{cases}$$ asymptote equation

The charcteristic equ.

$$1 + \frac{k}{s(s+1)(s+2)} = 0$$

$$\Rightarrow \frac{s(s+1)(s+2)+k}{s(s+1)(s+2)} = 0 \Rightarrow k = -s(s+1)(s+2)$$

$$\frac{dk}{ds} = 0 \Rightarrow 3s^2 + 6s + 2 = 0 \Rightarrow \begin{cases} s_1 = -0,42 \checkmark \\ s_2 = -1,57 \ \alpha \end{cases}$$ s_1 is acceptable between 0 and -1

$$\begin{cases} k(s_1) > 0 \ \checkmark \\ k(s_2) < 0 \ \times \end{cases}$$

Intersection with jw axis

$$s^3 + 3s^2 + 2s + k = 0 \quad k > 0$$

s^3	1	2
s^2	3	k
s	$\frac{6-k}{3}$	0
s^0	k	

$6 - k = 0 \Rightarrow k = 0$

Auxilary equation

$3s^2 + k = 0$

Roots of this equation are some of the roots for the characteristic equation

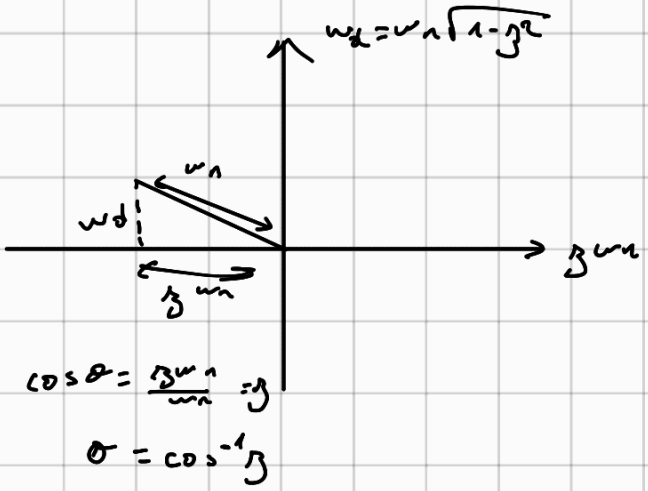
$3s^2 + 6 = 0 \Rightarrow s^2 = -2 \Rightarrow s_{1,2} = \pm j\sqrt{2}$

$k < 6$ $\quad 0 \le k \le 6$

$\zeta = 0,5$ $\quad \theta = \cos^{-1}\zeta = \cos^{-1} 0,5$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\zeta = ?$



$\cos\theta = \frac{\zeta\omega_n}{\omega_n} = \zeta$

$\theta = \cos^{-1}\zeta$

$$\frac{k}{s(s+1)(s+2)}$$

$s = \sigma + j\omega$

$G(s) = -\angle s - \angle s+1 - \angle s+2 = 180(2i+1)$

$= -\angle \sigma + j\omega - \angle \sigma + 1 + j\omega - \angle \sigma + 2 + j\omega =$

$= -\tan^{-1}\left(\frac{\omega}{\sigma}\right) - \tan^{-1}\left(\frac{\omega}{\sigma+1}\right) - \tan^{-1}\left(\frac{\omega}{\sigma+2}\right) = 180(2i+1)$

$\Rightarrow -\tan^{-1}\left(\frac{\omega}{\sigma}\right) - \tan^{-1}\left(\frac{\omega}{\sigma+1}\right) = +\tan^{-1}\left(\frac{\omega}{\sigma+2}\right) + 180(2i+1)$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

$$-\tan\left(\tan^{-1}\left(\frac{\omega}{\sigma}\right) + \tan^{-1}\left(\frac{\omega}{\sigma+1}\right)\right)$$

$$1 - \tan\left(\tan^{-1}\left(\frac{\omega}{\sigma}\right)\right)\tan\left(\tan^{-1}\left(\frac{\omega}{\sigma+1}\right)\right)$$

$$= \frac{\tan\left(-\tan^{-1}\left(\frac{\omega}{\sigma+2}\right)\right) + 0}{1 - \tan\left(\tan^{-1}\left(\frac{\omega}{\sigma+2}\right)\right) \times 0}$$

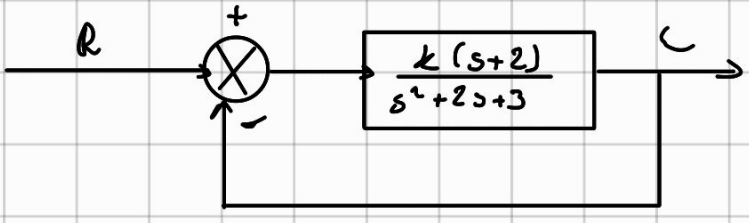
$$\frac{\frac{\omega}{\sigma} + \frac{\omega}{\sigma+1}}{1 - \left(\frac{\omega}{\sigma}\right)\left(\frac{\omega}{\sigma+1}\right)} = \frac{\omega}{\sigma+2}$$

β $\qquad$ β

$$\frac{k\,G(s)}{1+k\,G(s)\,H(s)} = 0$$

$$|G(s)\,H(s)| = \frac{1}{k}$$

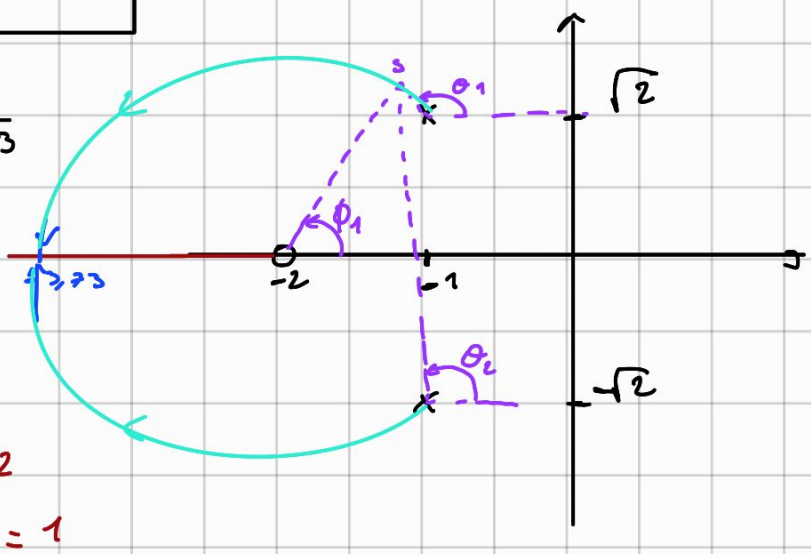
checking the magnitude condition at $s = s_1$ or s_2



$$G(s) = OLTF = \frac{k(s+2)}{s^2+2s+3}$$

$z_1 = -2$

$p_{1,2} = -1 \mp j\sqrt{2}$



n = number of poles = 2

m = " " zeroes = 1

$$\text{centroid} = \frac{\Sigma \text{ real of poles} - \Sigma \text{ real of zero}}{1}$$

$$= \frac{-1-1-(-2)=0}{1}$$

break in point:

$$1 + \frac{k(s+2)}{s^2+2s+3} = 0$$

$$k = -\frac{s^2+2s+3}{s+2}$$

$$\frac{dk}{ds} = 0 \Rightarrow \checkmark \begin{cases} s_1 = -0.268 \times \\ s_2 = -3.732 \quad \checkmark \text{ acceptible} \end{cases}$$

$$\phi_1 - \theta_1 - \theta_2 = 180$$

$$-\theta_1 = 180 - \phi_1 + \theta_2$$

Departure angles from complex poles

$\theta_1 = 180 - 90 + 55.145^\circ$

for the complex conjugate pair the departure angle is 145°

$$\angle s+2 - \angle s^2+2s+3 = 180(2i+1)$$

$s = \sigma + j\omega$

$$\angle s+2 - \angle s+1-j\sqrt{2} - \angle s+1\,j\sqrt{2} = 180(2i+1)$$

$$\tan^{-1}\left(\frac{\omega}{\sigma+2}\right) - \tan^{-1}\left(\frac{\omega-\sqrt{2}}{\sigma+1}\right) - \tan^{-1}\left(\frac{\omega+\sqrt{2}}{\sigma+1}\right) = 180(2i+1)$$

$$\tan\left(\tan^{-1}\left(\frac{\omega}{\sigma+2}\right) - \tan^{-1}\left(\frac{\omega-\sqrt{2}}{\sigma+1}\right) = \tan^{-1}\left(\frac{\omega+\sqrt{2}}{\sigma+2}\right) + 180(2i+1)\right]$$

$$\frac{2\omega(\sigma+1)}{(\sigma+1)^2-(\omega^2-2)} = \frac{\omega}{\sigma+2} \Rightarrow \omega\left[(\sigma+2)^2+\omega^2-3\right]=0$$

if $\omega = 0$

if $(\sigma+2)^2+\omega^2 = (\sqrt{3})^2$

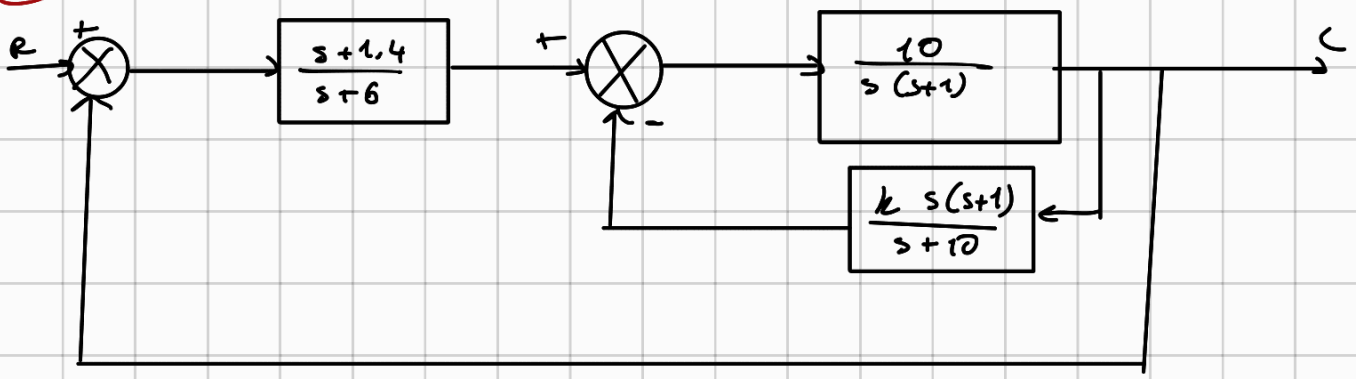
$(-2,0)$ Radius $=\sqrt{3}$

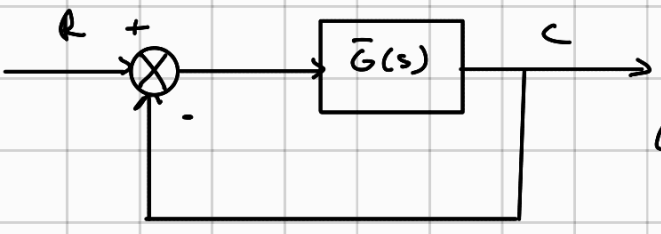
and $(x-x_0)^2+(y-y_0)^2=R^2$

circle

(x_0, y_0), radius R

Example





$$\frac{k\,G(s)}{1+k\,G(s)\,H(s)}$$

$$\bar{G}(s) = \left(\frac{s+1.4}{s+6}\right)\left(\frac{10(s+10)}{s(s+1)(s+10+10k)}\right)$$

Characteristic eq

$$1 + \bar{G}(s) = 0$$

$$\frac{(s+6)\, s(s+1)(s+10+10k) + (s+1,4)\, 10(s+20)}{\Phi} = 0$$

$$\underbrace{s(s+1)(s+6)(s+10)}_{\alpha_1} + 10ks(s+1)(s+6) + \underbrace{10(s+1,4)(s+20)(s+10)=0}_{\alpha_2}$$

$$1 + k\,\frac{10\, s(s+1)(s+6)}{s(s+1)(s+6)(s+20) + 10(s+1,4)(s+20)} = 0$$

$z_1 = 0 \quad z_2 = -1 \quad z_3 = -6$

$p_1 = -10, \quad p_2 = -3,69, \quad p_{3,4} = -1,65 \pm 1,07\,j$

